Name:_____

Exam Question Booklets
OCR Physics A

Booklet 1
10 Questions
69 marks

Things to remember:

✓

✓

✓

✓

✓

✓

✓

1.

A unit is often expressed with a prefix. For example, the gram may be written with the prefix 'kilo' as the kilogram. The prefix represents a power-of-ten. In this case, the power-of-ten is 10^3.

Complete Fig. 1.1 to show each prefix with its symbol and power-of-ten.

prefix	symbol	power-of-ten
kilo	k	10^3
nano	n	.....
centi	.....	10^{-2}
.....	M	10^6
.....	T	10^{12}

Fig. 1.1

[4]

2.

(a) A plane has an air speed of 240 km h^{-1} due north. A wind is blowing at 90 km h^{-1} from east to west. Use a vector triangle to calculate the resultant velocity of the plane.

velocity = ... m s^{-1},

direction with respect to north = ... °

[4]

(b) The plane flies under these conditions for 10 minutes. Calculate the component of the displacement

1. due north,

displacement = ... km

2. due west.

displacement = ... km
[2]

3.

(a) State the difference between a vector and a scalar quantity.

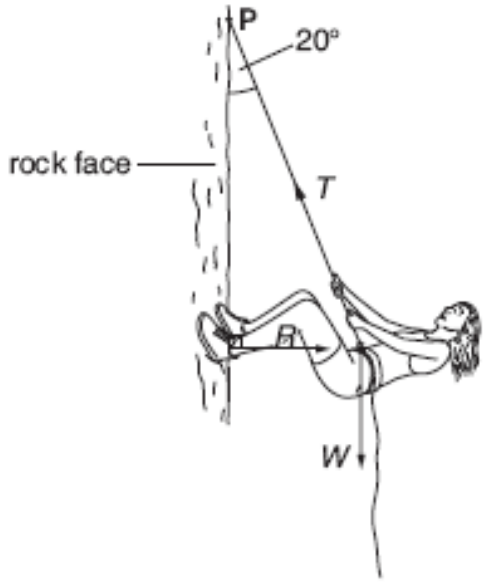
.....

..... [1]

(b) In the following list underline **all** the scalar quantities.

displacement kinetic energy mass power velocity weight [1]

(c) Fig. 1.1 shows a climber on a vertical rock face supported by a rope. The climber is in equilibrium.



The weight of the climber and her equipment is 650 N. The rope is attached to the climber and fixed to a point **P** where it makes an angle of 20° to the vertical. The contact force R acts on the climber at right angles to the rock face.

(i) Use a vector triangle or resolve forces to calculate

1 the tension T in the rope

T = ... N

2 the contact force R.

R = ... N

[3]

(ii) The climber moves down the rock face and the angle the rope makes with the vertical decreases. Explain why the magnitude of the tension decreases.

.....

.....

..... **[1]**

[Total: 6]

4.

(a) Fig. 1.1 shows a table of vector and scalar quantities.

speed, acceleration	
energy, power	
force, pressure	
velocity, displacement	

Fig. 1.1

In the blank spaces provided in Fig. 1.1, label the pair of quantities that are both vectors with a V and the pair that are both scalars with an S. [2]

(b) Fig. 1.2 shows the path taken by an athlete when she runs a 200 m race in 24 s from the start position at S to the finish at F.

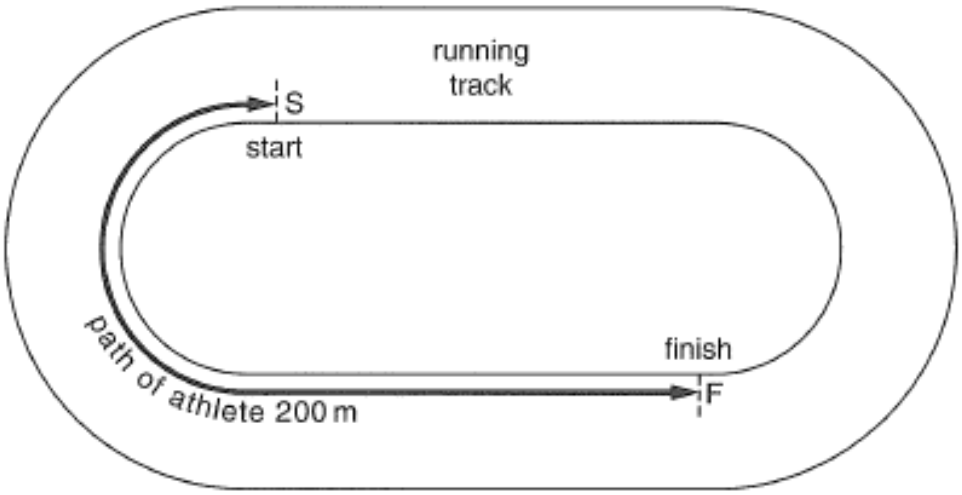


Fig. 1.2

(i) Calculate the average speed of the athlete.

average speed = ...m s^{-1} [2]

(ii) Explain how the magnitude of the average velocity of the athlete would differ from her average speed. A quantitative answer is not required.

.....

.....

.....

.....[2]

[Total: 6]

5. **(a)** **(i)** Define *speed*.

.....

.....

(ii) Define *velocity*.

.....

.....

(iii) State and explain the differences between these quantities.

.....

.....

.....

[4]

(b) Fig. 1.1 shows a fairground big wheel. The wheel is rotating in a vertical plane and carriages travel round a circle of diameter 40 m at a constant speed. The carriages complete one revolution in 3.5 minutes.

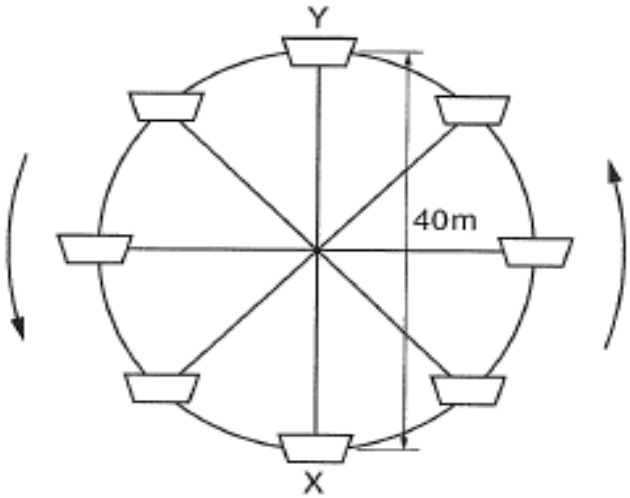


Fig. 1.1

(i) A carriage moves half a revolution from X to Y. Calculate

1. the speed of the carriage

speed = ... $m\,s^{-1}$

2. the magnitude of the average velocity of the carriage.

magnitude of the average velocity = ... $m\,s^{-1}$
[4]

(ii) The carriage in **(b)(i)** returns to point X. Calculate, for the **complete revolution**,

1. the speed of the carriage

speed = ... $m\,s^{-1}$

2. the average velocity of the carriage.

average velocity = ... $m\,s^{-1}$

Comment on your answer.

.....

..... [3]

(c) Describe how the instantaneous velocity of the carriage at Y differs from the average velocity of the carriage after travelling from X to Y.

.....

.....

.....

..... [3]
[Total: 14]

6.

(a) Distinguish between scalar and vector quantities.

.....

.....[2]

(b) Fig. 1.1 lists a number of physical quantities. Put a (✓) in the box next to the quantities that are vectors.

acceleration	
density	
energy	
momentum	
power	
velocity	

Fig. 1.1 [3]

(c) Fig. 1.2 shows a force of 6.0 N acting at 30° to the horizontal.

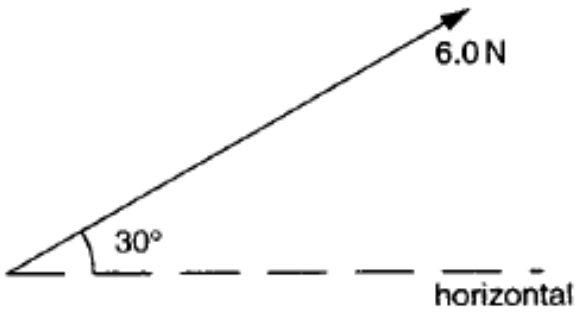


Fig. 1.2

Calculate the component of the force that acts

(i) horizontally,

horizontal component ..... N

(ii) vertically.

vertical component ..... N

[2]

7. **(a)** Define *acceleration*.

.....

..... [1]

(b) Fig. 3.1 shows the variation of the velocity v, with the time t, of a train as it travels from one station to the next. The mass of the train is 105 tonnes.

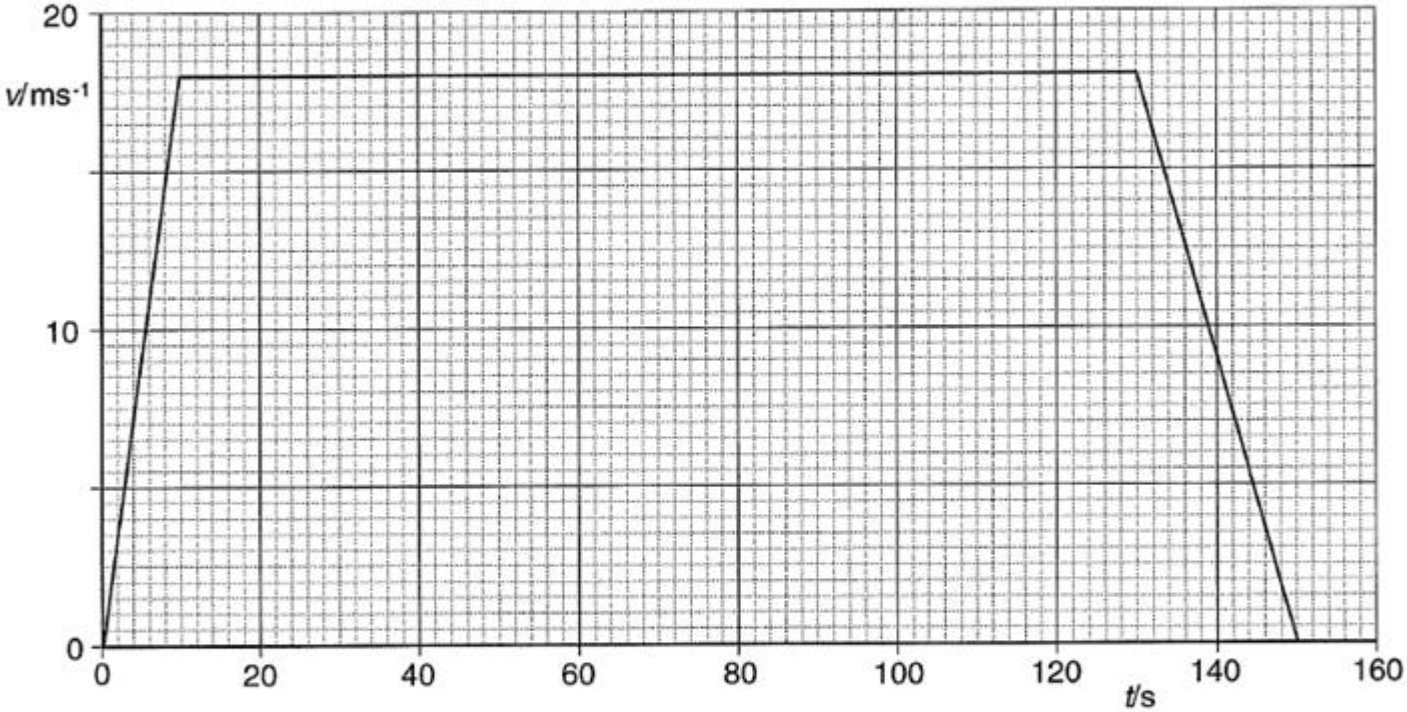


Fig. 3.1

(i) Use Fig. 3.1 to calculate the acceleration of the train in the first 10.0 s.

acceleration = ... $m\,s^{-2}$ [2]

(ii) Use Fig. 3.1 to calculate the distance between the two stations.

distance = ... m [3]

8.

A girl falls vertically onto a trampoline, as shown in Fig. 2.2.

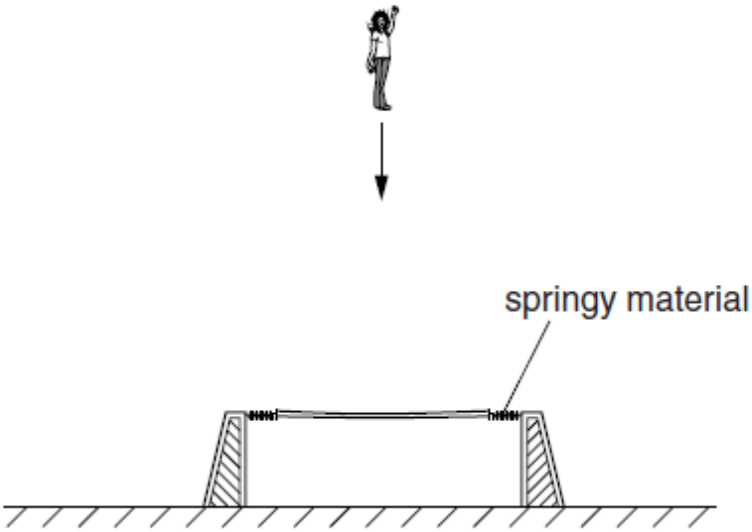


Fig. 2.2

The trampoline consists of a central section supported by springy material. At time $t = 0$ the girl starts to fall. The girl hits the trampoline and rebounds vertically. The variation with time t of velocity v of the girl is illustrated in Fig. 2.3.

For the motion of the girl, calculate

(i) the distance fallen between time $t = 0$ and when she hits the trampoline,

distance = ... m [2]

(ii) the average acceleration during the rebound.

acceleration = ... $m\,s^{-2}$ [2]

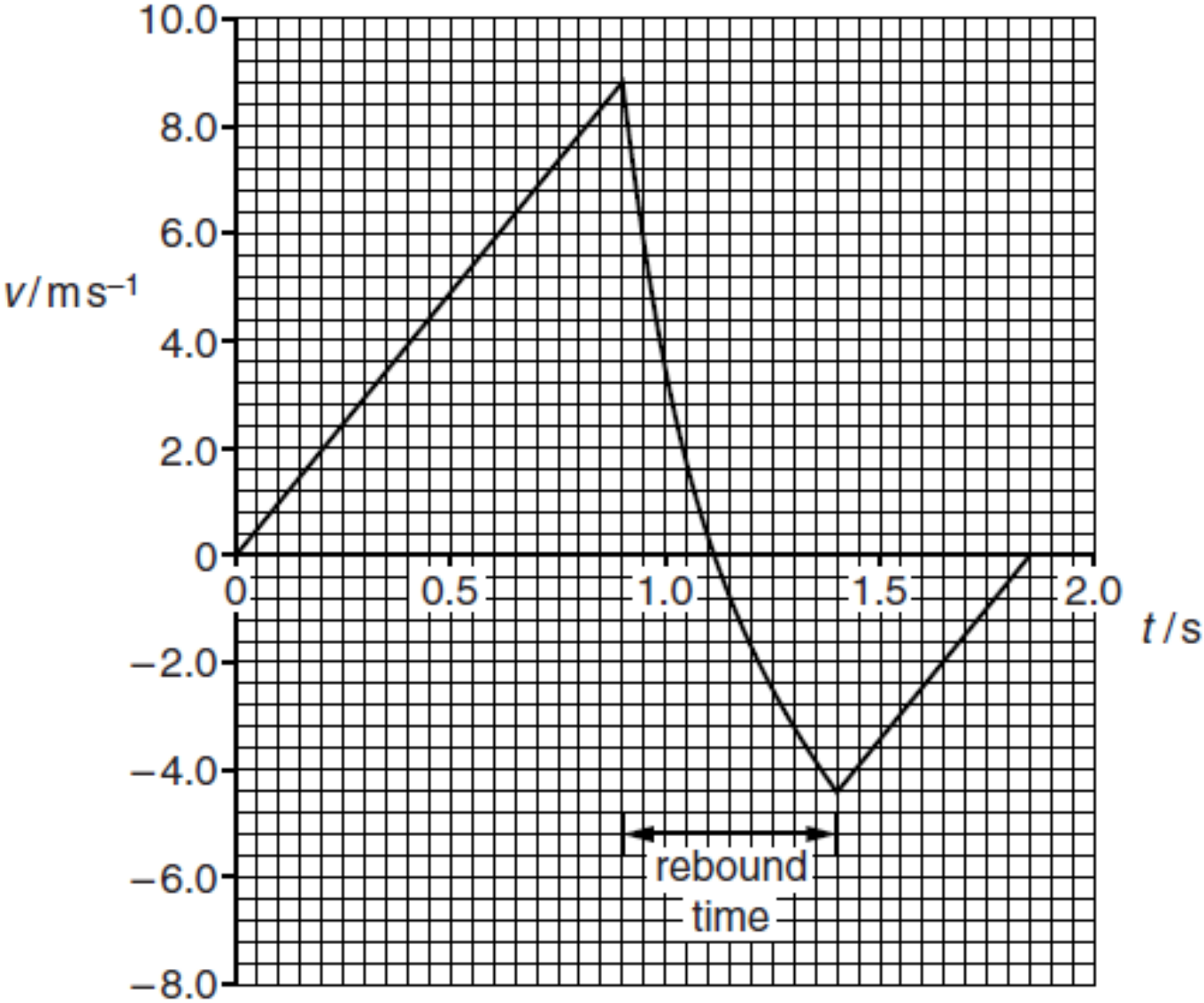


Fig. 2.3

(iii)

Use Fig. 2.3 to compare, without calculation, the accelerations of the girl before and after the rebound. Explain your answer.

.....

.....

.....[2]

9.

(a) Fig. 2.1 shows a graph of velocity v against time t for a train that stops at a station.

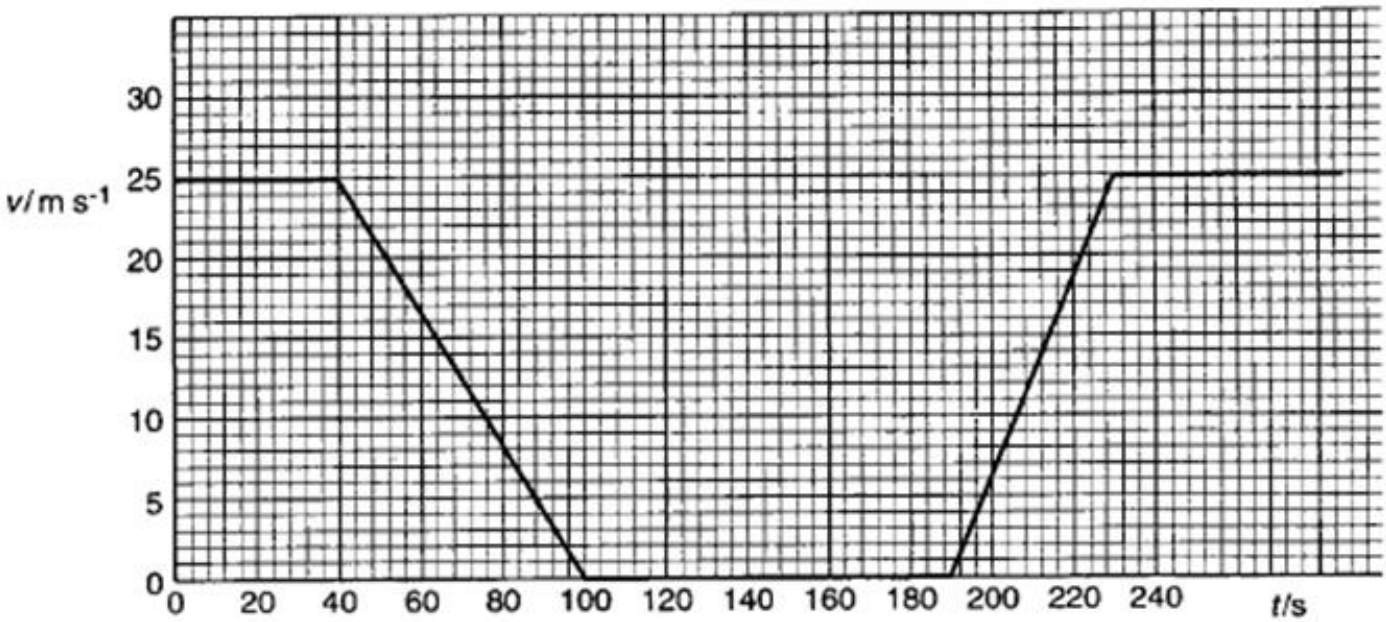


Fig. 2.1

(i) For the time interval $t = 40$ s to $t = 100$ s, calculate

1. the acceleration of the train,

acceleration ..... m s^{-2}

2. the distance travelled by the train.

distance ..... m

[5]

(ii) Calculate the distance travelled by the train during its acceleration from rest to 25 m s^{-1}.

distance ..... m [2]

(iii) Calculate the journey time that would be saved if the train did not stop at the station but continued at a constant speed of $25\,m\,s^{-1}$.

time saved ..... s [4]

10.

(a) Fig. 4.1 shows a graph of the velocity v against time t for a moving object.

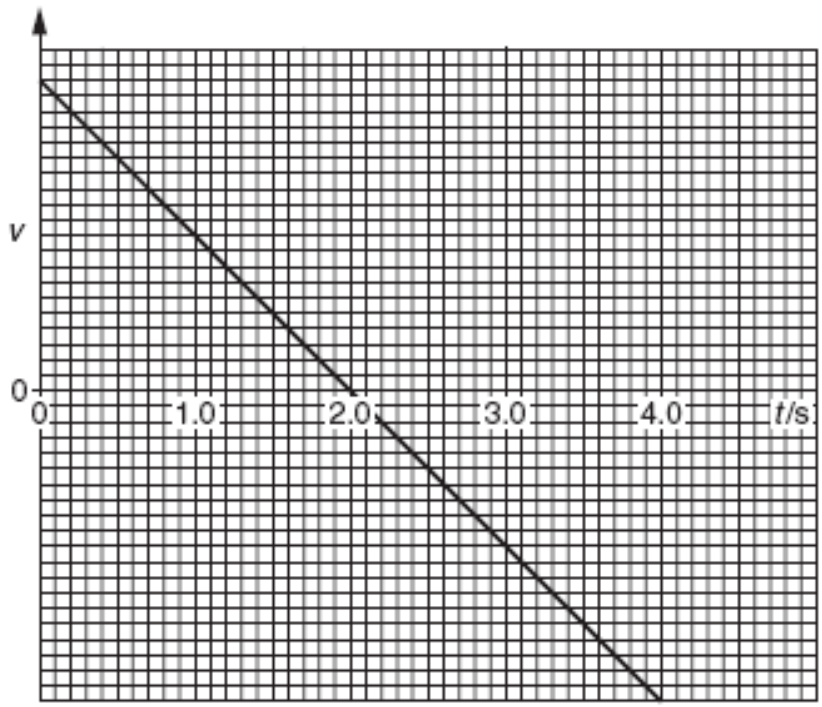


Fig. 4.1

Use the graph to describe and explain how the velocity and acceleration of the object change with time. Use the values on the time axis as reference for your description.

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.....

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.....

.....

..... [3]